ENERGY SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military Action Extension Methodology

DBA-EN-SF: Energy Sector Parent Document

Governing Parent for DBA-ES, DBA-GAS, DBA-OIL,
and All Sub-Sector Instruments in the Energy Sector Series

Version 0.19 — DRAFT

November 2026

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For review by engineers, regulators, asset owners, and policy analysts responsible for protecting energy sector critical infrastructure across the electric, natural gas, and petroleum sub-sectors.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the NIPP Energy sector. It is the parent instrument for three sub-sector series: DBA-ES (Electric sector), DBA-GAS (Natural Gas sector), and DBA-OIL (Petroleum sector). Each sub-sector series inherits the methodology and organizing principles established here and applies them to the specific facility types, operational characteristics, and regulatory contexts of its sector.

The Energy sector, as defined by the National Infrastructure Protection Plan (NIPP), encompasses the electric power system, natural gas production and delivery infrastructure, and petroleum production and refining infrastructure. DHS is the Sector-Specific Agency (SSA) for the Energy sector, with DOE, FERC, NERC, TSA, and PHMSA as the primary federal regulatory and operational partners depending on sub-sector and function.

This document applies the DB→DBA→DBA-MA three-layer analytical progression at the Energy sector level: a baseline Design Basis (DB) establishing the postulated conditions the sector's facilities must withstand; a Design Basis Accident (DBA) layer bounding the non-adversarial consequence envelope; and a military action extension (DBA-MA) applying the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to the sector's governing postulated events. The sector-level instruments establish this progression as a requirement; each sub-sector framework and facility-class instrument develops it in detail.

1.2 What This Document Does Not Do

This document does not supersede or modify any existing sub-sector framework or facility-class instrument. DBA-ES, DBA-GAS, and DBA-OIL are established series with their own governing documents; this parent establishes the common architecture they share and the cross-sector cascade analysis that spans their boundaries.

This document does not rename the sub-sector series. DBA-ES, DBA-GAS, and DBA-OIL retain their existing designations. The DBA-ES series retains its internal naming convention (DBA-ES-HY-FC1, DBA-ES-TX-FC2, etc.) without modification.

This document does not create new facility class designations. FC designations are the authority of their respective sub-sector frameworks. This document carries a series of inventory tables that reflect the current FC register state across all three sub-sectors.

1.3 Inductive Architecture Principle

The Energy sector series is built inductively: parent documents emerge from mature children, not the reverse. This document was first drafted at v0.3 because the three sub-sector frameworks had

by then reached sufficient maturity to reveal the common architecture they share. This document does not direct the sub-sector frameworks; it reflects and codifies what they have established.

Future revisions of DBA-EN-SF will be triggered by substantive developments in the sub-sector frameworks — new facility classes, new cross-sector cascade methodologies, or changes to the governing regulatory context — not by a top-down revision schedule.

1.4 Sub-Sector Series Inventory

This section held the series inventory until 9 September 2026. The authoritative version state of the series is now DBA-EN-REG, the Energy Sector Series Register, a standalone instrument issued that day: a register whose members move independently of one another does not live inside one of them, and the inventory here had twice gone stale because recording a child's increment meant opening this parent. Every inventory or version statement in the series reconciles to DBA-EN-REG. This section states no versions of its own, and the table it formerly carried is retained in v0.17 in Project 8's archive.

Section 2 — Energy Sector Taxonomy

2.1 Organizing Framework

The Energy sector's facility taxonomy is organized into two primary branches corresponding to the NIPP Energy sector structure: Electricity and Oil and Natural Gas. The Electricity branch covers generation (GenCo), transmission (TransCo), distribution (DisCo), and the control-company function (CntlCo). The Oil and Natural Gas branch covers crude oil production, midstream, and petroleum processing; and natural gas production, transmission, distribution, LNG, and storage.

The taxonomy in this section is the authoritative reference for series-wide facility scoping decisions. When a new facility class is proposed for any sub-sector series, its placement — which sub-sector series it belongs to, which consequence chain governs its inclusion, and which regulatory anchor applies — is determined against this taxonomy. The governing principle is consequence-chain placement: a facility belongs in the sub-sector series whose primary consequence chain it affects most directly.

2.2 GenCo — Generation by Type

The generation taxonomy spans seven primary generation types. Nuclear generation is addressed in the DBA-MA nuclear series and is referenced here for completeness; its grid consequence chain (Zaporizhzhia: loss of Europe's largest nuclear plant as both a nuclear safety event and a grid reliability event) is addressed in the DBA-ES series and the DBA-MA Introduction.

Generation Type	Sub-Types	Primary Regulators	DBA-EN Series Treatment	FC Status

Fossil fuel	Coal, natural gas, oil-fired	TSA (security); NERC/FERC (reliability); State environmental (emissions)	Gas-fired generation is addressed through a gas-electric cascade in DBA-ES-GC-FC4 and DBA-GAS-FC1. Coal: DBA-ES-GN-FC6. Oil-fired: DBA-ES-GN-FC7.	FC4 (gas), FC6 (coal), FC7 (oil-fired)
Nuclear	Large LWR (PWR, BWR), SMR, VVER	NRC (US); STUK (Finland); SNRIU (Ukraine); IAEA (international)	Addressed in the DBA-MA nuclear series. Grid consequence chain (Zaporizhzhia) addressed in DBA-ES and DBA-MA Introduction §6.3.	DBA-MA series
Hydropower	Large storage (>500 MW), run-of-river, pumped storage, tidal	FERC Part 12; NERC CIP-014; USCOLD; Army Corps of Engineers	Large hydroelectric storage is addressed in DBA-ES-HY-FC1. Run-of-river and pumped storage: deferred FC. Tidal: not yet scoped.	DBA-ES-HY-FC1
Solar	Utility-scale PV, solar thermal (CSP), distributed rooftop	FERC; NERC (interconnection); State PUC (distributed)	Utility-scale solar at cluster level: deferred FC. Distributed solar is addressed as an edge generation component in DBA-ES-DI-FC5.	Partial (FC5)
Wind	Onshore utility-scale, offshore, distributed	FERC; NERC; BOEM (offshore); State PUC	Wind at cluster level: deferred FC. Distributed wind is addressed as an edge generation component in DBA-ES-DI-FC5. Offshore wind: DBA-ES-GN-FC8.	Partial (FC5, FC8)
Battery / BESS	Utility-scale BESS, distributed storage	FERC Order 841; NERC; State PUC	Utility-scale BESS: DBA-ES-GN-FC8. Distributed BESS is addressed as the edge generation component in DBA-ES-DI-FC5.	FC8, partial (FC5)

Geothermal, biomass, wave, tidal	Various	FERC; State PUC; applicable environmental agencies	Not yet scoped. Consequence chains have not yet been analyzed.	Not yet

2.3 TransCo — Transmission Owners and Operators

The transmission taxonomy covers the physical infrastructure connecting generation to load at bulk power system voltage levels. Voltage class specification and bus bar identifiers per circuit are the technical parameters that govern facility class placement within DBA-ES: the 345 kV threshold separates FC2 (EHV substations) from sub-transmission facilities outside the current series scope.

Infrastructure Element	Technical Specification	Regulatory Anchor	DBA-EN Series Treatment
EHV transmission substations	345–765 kV; metropolitan ring configuration; CIP-014 Tier 1/2 designation; LPT with RT_HVLLC ≥12 months	NERC CIP-014; FERC Order 693; TPL standards	DBA-ES-TX-FC2
HVDC interconnects	>2,000 MW; cross-interconnection role; LCC or VSC technology; point-to-point (primary scope) or multi-terminal (deferred)	FERC; NERC FAC standards; applicable TPL	DBA-ES-TX-FC3
Transmission lines	Voltage class per facility: 115 kV, 230 kV, 345 kV, 500 kV, 765 kV. Bus bar identifiers per circuit.	NERC FAC-001/002; FERC; State PUC	Deferred FC; addressed at the system level in TPL standards
SCADA and control communications	ICCP; SCADA RTUs; substation automation; protective relaying	NERC CIP-005/006/007; FERC Order 706	CE events in FC2, FC3; SL events in CntlCo instrument
Protective relay systems	Distance, differential, overcurrent, frequency, voltage	NERC PRC standards; IEEE relay standards	CE-2/CE-3 events in FC2; CE-3 in FC5; UFLS position paper

	protection; PRC-024 ride-through settings		

2.4 DisCo — Distribution Functions

The distribution taxonomy covers the infrastructure delivering power from the transmission system to end-use customers. The PMU (Phasor Measurement Unit) is specifically called out in the DisCo control systems because it is the primary measurement instrument underlying the UFLS frequency wavefront analysis in the Predictive UFLS Ride-Through Position Paper — the cross-reference between DisCo instrumentation and the UFLS paper is a series linkage that must be maintained.

Infrastructure Element	Technical Specification	Regulatory Anchor	DBA-EN Series Treatment
Distribution substations with edge generation	HV/EHV input (115 kV+); 69–110 kV output; microgrid or DER assets; islanding capability with Command Broker architecture	NERC CIP (where applicable); IEEE 1547; State PUC; FERC Order 2222	DBA-ES-DI-FC5
Loads	Commercial, industrial, residential. Load classification: firm/fixed vs. sheddable. Defense-critical loads: DSB finding (31 of 34 DoD most critical assets on commercial DisCo infrastructure).	State PUC; FERC (wholesale); DOD (defense-critical)	XC events in FC2, FC5; Tier 3 acceptance criteria
Distribution control systems	SCADA (2–5 second polling cycle); DCS; PLC; EMS (Energy Management System); PMU (Phasor Measurement Unit — wavefront analysis; cross-reference Predictive UFLS Ride-Through Position Paper)	NERC CIP; State PUC; IEEE standards	CE events in FC5; PMU data cited in UFLS position paper

Electricity markets	Wholesale market dispatch; retail rate structures; demand response programs; ancillary services	FERC (wholesale); State PUC (retail)	Market timing dependency in XC-4 events (FC3, FC4)

2.5 CntlCo — Balancing Authority, Reliability Coordinator, Transmission Operator

The CntlCo taxonomy is more tool-specific than the physical infrastructure taxonomies because the CntlCo's design basis is anchored in its software-data layer as much as its physical and human layers. The RC and BA tool suites, the NERC CIP compliance registry classification scheme, and the primary and backup control center communications architecture are all design basis elements addressed in the DBA-ES-CntlCo coordination class instrument.

Function / Tool	Description	NERC Registration	DBA-EN Series Treatment
Reliability Coordinator (RC) tools	SCADA; Electronic Tagging System; Telephone System; OATI Suite of Applications; State Estimator; EPRI TagNet; Power World; RCIS (Reliability Coordinator Information System)	RC	SL events in DBA-ES-CntlCo §5; RC extension §7.2; SABD metric governs state estimator loss consequence
Balancing Authority (BA) tools	Transmission Energy Scheduler; Eastern Interconnect Frequency Monitor; Security Constrained Unit Commitment; Telephone System; SCADA; RCIS; Energy Management System (EMS)	BA	SL events in DBA-ES-CntlCo §5; BA primary case §2; SABD governing metric
NERC CIP compliance registry	Critical cyber asset classification: BCA (BES Cyber Asset); EACMS (Electronic Access Control and Monitoring System); PACS (Physical Access Control System); PCA (Protected Cyber Asset). Impact	All CIP-registered entities	CE-4 (cyber intrusion) in FC2; SL-5 in CntlCo; governs CIP-014 designation trigger for FC2 scope

	classification: High / Medium / Low.		
Primary and backup control center communications	Data communications; Voice; Radio; Telephone System; SCADA; ICCP. Primary and BCC must be geographically independent per DBA-ES-CntlCo acceptance criteria.	BA, RC, TOP	PL-4, PL-5 events in DBA-ES-CntlCo; BCC geographic independence acceptance criterion
Transmission Operator (TOP)	Operates transmission facilities in real time; implements RC and BA directives; controls switching at FC2 and FC3 facilities; must maintain manual field switching capability under SCADA loss	TOP	TOP extension in DBA-ES-CntlCo §7.3

2.6 Oil and Natural Gas

The Oil and Natural Gas branch of the energy taxonomy covers two commodity streams with partially overlapping infrastructure. The governing cross-sector cascade — the gas-electric bidirectional feedback loop — is addressed in Section 3. The taxonomy below covers the upstream, midstream, downstream, and storage segments of each commodity stream.

Segment	Infrastructure	Technical / Operational Notes	Primary Regulators	DBA-EN Series Treatment
Crude Oil — Upstream	Onshore fields; offshore fields; gathering systems	Wellhead pressure management, produced water handling, and gathering line integrity	PHMSA; EPA; State environmental; BSEE (offshore)	DBA-OIL-FC2 for offshore production. Onshore upstream: not yet scoped.
Crude Oil — Midstream	Terminals; pipelines; storage (tank farms)	Crude transport by pipeline, rail, and marine. Tank farm fire and explosion risk. Colonial Pipeline paradigm	PHMSA; TSA; FERC; EPA	DBA-OIL-FC3 for terminals and storage. DBA-OIL-FC4 for transport pipelines. SDD metric governs

		applicable.		transport consequence.
Petroleum Processing	Refineries, blending terminals, petrochemical plants	Process unit fire and explosion as the governing SE event. Long lead time for major process units. RT_HVLLC concept applicable.	TSA; EPA; OSHA; State environmental	DBA-OIL-FC1
Natural Gas — Upstream	Onshore fields; offshore fields; gathering systems; processing plants	Gas processing separates NGLs from methane. Processing plant disruption reduces the pipeline quality gas supply.	PHMSA; FERC; EPA; BSEE (offshore)	Not yet scoped in DBA-GAS series
Natural Gas — Transmission	Interstate and intrastate pipelines; compressor stations; metering and regulation (M&R) stations	Compressor stations in critical GenCo supply corridors are the governing FC4 facility class. GSD metric governs electric consequence.	FERC; PHMSA; TSA (security directives)	DBA-GAS-FC1; DBA-ES-GC-FC4
Natural Gas — Distribution	Local distribution companies (LDCs), service laterals, and customer meters	Residential heating dependency: dual-use pipeline (generation fuel + heating fuel) produces compound XC-4 consequence during winter peak.	State PUC; PHMSA	DBA-GAS-FC4; XC-4 in DBA-ES-GC-FC4
LNG Facilities	Liquefaction terminals; storage tanks; regasification terminals; import/export	LNG as a backup supply for pipeline disruption. Sabotage of LNG import terminal removes backup	FERC; PHMSA; USCG; DOE (export)	DBA-GAS-FC-LNG (annex DBA-GAS-UC-LNG)

	terminals	gas supply during extended pipeline outage.		
Storage	Underground storage (depleted reservoirs, aquifer, salt cavern); above-ground LNG storage	Strategic storage buffer: days to weeks of deliverability. Storage disruption extends GSD duration beyond compressor station recovery.	FERC; PHMSA; State	DBA-GAS-FC3

Section 3 — Cross-Sub-Sector Postulated Events

3.1 Status as a First-Class Postulated Event

The gas-electric bidirectional cascade is established in this document as a first-class cross-sub-sector postulated event at the DBA-EN level. It is not a secondary consequence or a beyond-design-basis contingency: it is a primary postulated event that must be analyzed in both the DBA-ES series (from the electric system's perspective) and the DBA-GAS series (from the gas system's perspective). The two analyses are complementary; neither substitutes for the other.

The bidirectional character of the cascade — gas disruption reduces electric generation; electric disruption reduces gas compression — was established in DBA-ES-Sector_Framework §3.5 and is carried through every relevant FC instrument. Winter Storm Uri (February 2021) demonstrated both directions simultaneously and empirically. An adversary who understands the feedback structure can initiate the cascade from whichever end is more accessible and rely on the bidirectional dependency to amplify the consequence without further action.

3.2 Cascade Analysis Table

Cascade Direction	Mechanism	Governing Metric	Primary DBA-EN Instrument
Gas → Electricity	Natural gas compressor station loss reduces pipeline delivery pressure below the GenCo minimum inlet threshold. Gas-fired generation curtails output or trips offline. The electric system loses generation capacity. Approximately 40% of US electric generation is gas-fired; gas transmission disruption in a critical corridor reduces generation capacity at the electric system level.	GSD (Generation Supply Duration): continuous period of full gas-fired generation curtailment due to insufficient pipeline pressure. Defined in DBA-ES-GC-FC4 §1.3.	DBA-ES-GC-FC4; DBA-GAS-FC1
Electricity → Gas	Electric supply loss to electrically driven compressor stations reduces compression capacity. Pipeline pressure in the supply corridor declines. Gas supply to gas-fired generation and heating customers is curtailed. Self-reinforcing: reduced gas supply further reduces gas-fired generation, further reducing	GSD (reverse cascade): same metric applied to the electrically-initiated cascade. TCI (Transfer Capability Isolation, from DBA-ES-TX-FC3) may apply if	DBA-ES-GC-FC4 (XC-2); DBA-ES-TX-FC2; DBA-ES-TX-FC3

	electric supply to compressor station auxiliaries.	HVDC or transmission loss initiates the electric supply disruption.	
Compound (bidirectional)	Winter Storm Uri (February 2021) demonstrated the full bidirectional cascade: extreme cold simultaneously increased electric demand and reduced gas-fired generation supply by disabling compressor stations and gas wells. The compound event eliminated both directions of the cascade mitigation simultaneously. An adversary can replicate this compound cascade deliberately through coordinated timing.	Compound GSD + load curtailment. XC-4 event in DBA-ES-GC-FC4: winter peak demand timing maximizes both heating fuel and electric generation consequences.	DBA-ES-GC-FC4 (XC-4); cross-reference DBA-GAS-FC1

3.3 Cross-Sub-Sector Coordination Requirement

The gas-electric cascade is the primary reason DBA-EN-SF exists as a document distinct from the sub-sector frameworks. Each sub-sector framework addresses the cascade from its own consequence perspective: DBA-ES instruments analyze the electric consequence of gas infrastructure failure; DBA-GAS instruments analyze the gas infrastructure consequence of electric supply failure. The DBA-EN-SF level is where the full bidirectional cascade is formally established as a postulated event category that both sub-sector analyses must address.

Acceptance criteria assignment. A facility may fall within the scope of two DBA-EN series instruments at once. A compressor station in a critical GenCo supply corridor is within both DBA-ES-GC-FC4 and DBA-GAS-FC1, and is the governing case. Where a facility is dual-scope, each instrument's acceptance criteria are binding for the consequence tier that instrument analyzes. Neither instrument's criteria govern the other's consequence chain, and neither is subordinate to the other at the shared facility. The two analyses are complementary and do not duplicate each other; a facility that satisfies one instrument's acceptance criteria has not thereby satisfied the other's.

This rule assigns acceptance criteria; it does not set their severity. Tightening of the criteria governing a supply relationship, where the facility is the sole energy supply path for a facility carrying a Tier 2 or Tier 3 consequence designation, is governed by the Cross-Sector Consequence Elevation Rule at §3.4. Assignment answers which instrument's criteria apply to which consequence. Elevation answers how tight those criteria must be. The two are independent: elevation tightens criteria that assignment has already placed with one instrument or the other, and assignment does not change because elevation has fired.

3.4 Cross-Sector Consequence Elevation Rule

The NERC gap this rule addresses. The North American Electric Reliability Corporation's CIP standards assign impact levels to bulk electric system cyber assets based on the electric-sector consequences of their loss. NERC CIP does not require, and has not established, any mechanism

by which the consequence designation of a facility in another sector elevates the design basis or impact classification of the energy sector asset that supplies it. The DBA-EN-SF cross-sector consequence elevation rule fills this gap as a pre-regulatory design basis requirement.

The rule. A DBA-EN series energy sector asset whose loss would deprive a facility carrying a Tier 2 or Tier 3 consequence designation in any DBA series instrument of its primary energy supply must carry tightened acceptance criteria for that supply relationship in its own DBA-EN design basis.

Trigger condition. The rule fires when two conditions are simultaneously satisfied. First, the dependent facility carries a Tier 2 or Tier 3 consequence designation in a DBA series instrument. Second, the energy sector asset is the sole necessary supply path: the dependent facility does not have an alternate energy supply of equivalent capacity. This single-path dependency criterion is drawn directly from the FC4 facility class definition (DBA-ES-GC-FC4 §1.2).

Self-referential trigger and series growth. The trigger is self-referential within the DBA series: the Tier 2 or Tier 3 designation of the dependent facility is established by its own DBA instrument, not by any external regulatory designation. This means the rule automatically expands as the DBA series matures. When the DBA-W series assigns a Tier 3 designation to a water treatment facility, every DBA-EN energy asset that is the sole supply path for that facility is immediately subject to the elevation rule.

Bidirectional implementation. The rule is implemented bidirectionally. The forward direction — the energy sector carries the design basis obligation imposed by dependent sectors' consequence tiers — is the foundational implementation established at v0.3. The reverse direction is now established for the named dependent series in the trigger table below. The European EPCIP / NIS2 framework is the governing international analog.

Named-series elevation trigger table:

Series	Triggering Tiers	Direction	Governing Instrument and Notes
DBA-W (Water Sector)	Tier 2 and Tier 3	Bidirectional. Forward: DBA-EN energy asset is the sole electric supply path for a Tier 2 or Tier 3 DBA-W facility. Reverse: DBA-W facility is the sole cooling water supply path for a Tier 2 or Tier 3 DBA-ES	DBA-EN-SF §5.4 (DBA-EN-SF / DBA-W Boundary Protocol). Full boundary protocol established at v0.4. Life-safety cascade anchored by dialysis center dependency (DBA-W §8.2)—instrument-level incorporation in DBA-W FC instruments: pending DBA-W series next revision.

		thermoelectric facility.	
DBA-DC (Data Center Series)	Tier 2 and Tier 3	Forward only. DBA-DC facilities are electricity consumers; no energy sector reverse dependency exists.	DBA-DC-L1 §5.2 and §5.3 establish the Tier 2 (grid-level frequency and voltage consequence) and Tier 3 (foundational digital dependency cascade) designations that trigger elevation. Instrument-level incorporation of the elevation obligation in DBA-DC FC instruments is an open item (OI-DC-1).
DBA-MA Nuclear Series	Tier 2 and Tier 3	Bidirectional. Forward: DBA-EN energy asset is the sole off-site power supply path for a nuclear facility; elevation implements the Station Blackout design basis (10 CFR 50.63). Reverse: nuclear facility loss is a Tier 2 or Tier 3 DBA-ES generation event.	Forward direction: NUC-001-4 (Nuclear Plant Interface Coordination) is the NERC implementing instrument. Instrument-level incorporation of the elevation obligation in DBA-MA nuclear FC instruments is an open item (OI-MA-1).

Notification protocol. When a DBA-EN series instrument is revised in a way that changes the consequence tier of a supply relationship governed by this rule, the revision history of that instrument must identify which cross-series instruments carry a dependency on that designation. Those instruments are flagged for review at their next scheduled revision.

Open items — instrument-level incorporation in dependent series. OI-DC-1: DBA-DC series instrument-level incorporation of the forward-direction elevation obligation. Resolution: DBA-DC-L1 next substantive revision. OI-MA-1: DBA-MA nuclear series instrument-level incorporation of the bidirectional elevation obligation. Resolution: DBA-MA US Regulatory Case next substantive revision.

3.5 The Coordinated Multi-Node Attack

The coordinated multi-node attack is established in this document as the second first-class cross-sub-sector postulated event at the DBA-EN level, beside the gas-electric cascade. Like the

cascade, it is a primary postulated event and not a beyond-design-basis contingency, and it applies to every sub-sector series: DBA-ES, DBA-GAS and DBA-OIL.

The event. An adversary selects the smallest set of facilities or elements whose loss, taken together, defeats the sector's ability to recover from the loss of any one of them, and removes that set within the window the system needs to recover. The set belongs to the system, not to any single facility in it. Each member may satisfy its own design basis in full and the set still succeeds. In the electric sub-sector it is the minimum target set of an interconnection, which for a load pocket served through a metropolitan ring may be as few as two substations. In natural gas it is the combination of compression, processing and storage losses that closes the simultaneity trap described in *Natural Gas Transmission* §5.2 (DBA-GAS-FC1). In petroleum it is a set of pipeline ruptures and pump-station losses that exceeds the capacity to repair them at the same time, as DBA-OIL-FC4 (*Crude and Product Transport*) establishes.

Its consequence. The event produces a Tier 2 or Tier 3 consequence from a small operational footprint. Where the set includes components with long replacement times, such as large power transformers, the consequence is more than a single outage. It becomes an extended regime, governed by the RT_HVLLC of the most constrained element in the set, during which the system runs with reduced capability for months.

Why single-contingency design is no defense. The energy sector is engineered to survive the loss of any one element. In the electric sub-sector this is the N-1 criterion. In gas and petroleum it is the sizing of storage, rerouting, mutual assistance and repair capacity against one event at a time. That design is calibrated to non-adversarial failure, which arrives one element at a time and at random. An adversary chooses which elements fail, when, and in what combination, and so chooses the one combination the criterion was never asked to survive. Compliance with a single-contingency criterion is therefore no evidence of protection against this event. Spacing the losses apart does not restore the defense either. An adversary who removes one element, waits for the system to re-optimize around the loss, and then removes the next finds a system that has already spent its margin. The sequential form exploits the recovery mechanism instead of outrunning it, and it demands less coordination than the simultaneous form.

Method and inheritance. DBA-EN-UC-2 (*Multi-Node Attack*) establishes the analytical method for identifying the set, the initiating condition classes, and the consequence trace for the electric interconnection and the gas-electric multi-corridor case. None of these is restated here. Each DBA-ES, DBA-GAS and DBA-OIL facility-class instrument states in its Military Action extension whether its class can be a member of such a set, and the outcome for that class, citing this section for the event. The requirement applies to all new instruments, and to existing instruments at their next substantive revision. The Cross-Sector Consequence Elevation Rule at §3.4 applies to this event as it does to the cascade. The two events also combine: an adversary can use a multi-node attack to strip the reserve margin that would otherwise contain the cascade (*Multi-Node Attack* §1.2).

Section 4 — Series-Wide Architectural Requirements

4.1 DB→DBA→DBA-MA Three-Layer Progression

All instruments in the Energy sector series — sub-sector frameworks, facility-class instruments, coordination class instruments, and use case documents — apply the DB→DBA→DBA-MA three-layer progression. The progression is non-negotiable: a facility-class instrument that addresses only the DBA layer without explicitly establishing the DB layer, or that addresses the DBA-MA extension without the DBA envelope, is architecturally incomplete.

Non-nuclear sector frameworks and facility-class instruments cannot assume reader familiarity with the DB→DBA→DBA-MA architecture and must state it explicitly. This requirement was established in DBA-ES-Sector_Framework and is adopted here as an Energy-sector-wide requirement.

4.2 Dual Design Basis Architecture

Consistent with DBA-ES §3.4 and the AI Governance series (Project 2), any energy sector facility may simultaneously carry a sector-specific physical Design Basis established by the applicable DBA-EN series instrument and an AI Governance Design Basis governing the deployment of ML and Gen AI systems in the facility's operational and control environment. The two design bases interact at the DBA level and must be developed in coordination.

Each sub-sector framework and facility-class instrument must acknowledge the dual-DB architecture and identify the interaction points between the physical DBA and the AI Governance DBA specific to its facility type. This requirement applies to all new instruments and to existing instruments on their next substantive revision pass.

4.3 Bright Line and Command Broker Architecture

The Bright Line and Command Broker architecture is a series-wide requirement for all energy sector instruments. The Bright Line is the Bright Line as defined in FD-BL (Foundational Definition — The Bright Line): a threshold on individual actions, not a property of a component, a technology class, or an architectural layer. Under FD-BL §4.2–§4.3, an action is above the line when its incorrect performance could produce a consequence at or above the facility's severity threshold and requires substantive, architecturally enforced authorization by a qualified Command Broker (FD-BL §4.5); an action is below the line when it could not. Whether a component is deterministic or generative does not decide placement — determinism is the scope carve-out of FD-BL §4.4, under which a component that is deterministic within a stated input bound and formally testable as deployed is outside the Command Broker obligation. Energy sector instruments do not restate this definition; they inherit it by reference to FD-BL and apply it to their facility types. Each sub-sector framework — DBA-ES, DBA-GAS, and DBA-OIL — and every facility-class and functional-class instrument beneath them takes the Bright Line as defined in FD-BL and develops only the sector-specific placement and Command Broker application for its facilities.

Sub-sector frameworks and facility-class instruments INHERIT the placement criterion BY REFERENCE to this section and SHALL NOT restate it in their own words. An instrument states the placement outcome for each of its interaction points and cites this section for the rule

that produced it. Restating the criterion per instrument is the drift mechanism recorded as OI-R3-1: fourteen instruments restated it, every restatement drifted from this section, and every one drifted the same way — toward placement by the character of the component rather than by the consequence of the action. This paragraph is the propagation closure for that defect class and is the reason the criterion is stated once, here.

The term “Bright Line” is used exclusively in the FD-BL sense throughout all DBA-EN series instruments. Any other use of the term in an energy sector instrument — for example, to mean federal coordination triggers or escalation thresholds — must be replaced with a distinct term. The applicable replacement term for federal coordination thresholds is “Federal Coordination Threshold.”

4.4 Consequence Metrics

Six consequence-duration metrics are established across the Energy sector series. Each is a series standard and must be used consistently in any instrument that addresses the relevant consequence type.

RT_HVLLC (Recovery Timeline for High-Value Long-Lead Components): the time required to restore a facility to service following destruction of its most difficult-to-replace major component, expressed in months. Established as a series standard in DBA-ES-Sector_Framework §6.2. Originated in GenMix Optimization Whitepaper (Roxey, Eclectic Technologies, 2026). Applies to all FC instruments where long-lead-time equipment is the governing consequence element.

TCI (Transfer Capability Isolation): the continuous duration during which cross-interconnection transfer capability is unavailable following loss or block of an HVDC link, expressed in hours, days, or months. Established in DBA-ES-TX-FC3 §1.3. Applies to HVDC facility class instruments.

GSD (Generation Supply Duration): the continuous period during which gas-fired generation capacity in an affected GenCo supply corridor is curtailed to zero due to insufficient pipeline delivery pressure, expressed in hours or days. Established in DBA-ES-GC-FC4 §1.3. Applies to gas compressor stations and gas transmission instruments where the governing consequence chain runs through the gas-electric cascade.

PPSD (Pipeline Pressure Supply Duration): the continuous period during which pipeline delivery pressure at the transmission system level is insufficient to maintain pipeline-quality gas supply to downstream compressor stations and GenCo facilities, following a major gas processing plant outage, expressed in hours or days. Established in DBA-GAS-FC2 §1.3 as the governing consequence metric for natural gas processing plant outage. PPSD is upstream of and sequential to GSD: PPSD measures the consequence of processing plant loss on pipeline pressure supply duration; GSD measures the downstream consequence of that pipeline pressure loss on gas-fired generation supply duration. The PPSD→GSD sequential relationship is formally established in DBA-EN-UC-1 §5.1. Applies to gas processing facility class instruments and to any cross-sector cascade analysis where processing plant loss is the upstream initiating condition.

SDD (Supply Delivery Duration): the continuous interval of pipeline delivery curtailment from the initiating event to restoration of full throughput capacity at the governing delivery point, expressed in hours or days. Established in DBA-OIL-FC4 v0.3 §§1.3 and 4.2 as the governing consequence metric for the crude and product pipeline transport facility class. SDD is measured at the governing delivery point — the terminal, refinery intake, or distribution hub whose curtailment produces the highest-tier consequence — not at the point of physical damage. SDD is not additive across pipeline segments: where multiple segments are affected, the governing SDD is the longest single-segment critical path, not the cumulative sum across segments. A Zone 1/2/3 acceptance criterion framework with T_1/T_2 calibration thresholds governs consequence tier assignment; site-specific calibration is required at implementation. Applies to crude and product pipeline transport facility class instruments and to cross-sector cascade analysis where liquid fuel delivery curtailment is the governing upstream consequence.

SABD (Situational Awareness Blackout Duration): the continuous interval between loss of state estimator validity and restoration of coordinated real-time operational authority at the Balancing Authority, Reliability Coordinator, or Transmission Operator level, expressed in minutes or hours and established in DBA-ES-CntlCo v0.2 §1.2 as the governing consequence metric for the CntlCo coordination class. SABD begins at the moment the state estimator is confirmed non-convergent, produces outputs flagged as unreliable, or is taken out of service due to suspected data integrity compromise. SABD ends at the moment a qualified system operator confirms the state estimator is producing convergent, validated results and resumes issuing dispatch instructions based on its output. Backup control center (BCC) transition does not automatically terminate SABD: a successful BCC transition that does not restore state estimator convergence leaves the SABD clock running at the BCC. A Zone 1/2/3 escalation structure with T_1/T_2 calibration thresholds governs consequence tier assignment; site-specific calibration is required. Applies to CntlCo coordination class instruments and to any cross-sector cascade analysis where situational awareness loss is a contributing condition.

4.5 Facility Class Naming Convention

The naming convention for Energy sector facility class instruments is: DBA-[SUB-SECTOR]-[TYPE CODE]-FC[NUMBER]. Type codes are assigned to distinguish facility classes within the same sub-sector by their governing infrastructure type. The DBA-DC series retains its existing designation without modification. Facility classes promoted from an existing class take the next sequential FC number (for example, DBA-OIL-FC5, carved from DBA-OIL-FC3); facility classes promoted from a named standalone use case may instead retain the use-case mnemonic in place of a sequential number (for example, DBA-GAS-FC-LNG, promoted from DBA-GAS-UC-LNG), preserving traceability to the originating instrument.

A sub-sector may further organize its facility classes into named subsector groupings — for example, the Critical Loads grouping (DBA-ES-CL) within the Electric sub-sector, which groups

load-side facility classes whose design basis is electric-sector-dominated and whose compromise elevates across sectors. Such groupings are organizational and do not alter the designation of the instruments they contain.

Designation	Sub-Sector	Type Code	Facility Class
DBA-EN-SF	Energy (parent)	SF = Sector Framework	This document — Energy sector parent
DBA-ES-Sector_Framework	Electric	SF = Sector Framework	Electric sub-sector parent
DBA-ES-HY-FC1	Electric	HY = Hydroelectric	Large hydroelectric storage (>500 MW)
DBA-ES-TX-FC2	Electric	TX = Transmission	Large EHV transmission substation (345–765 kV)
DBA-ES-TX-FC3	Electric	TX = Transmission	HVDC interconnect (>2,000 MW)
DBA-ES-GC-FC4	Electric	GC = Gas-electric Cascade	Major natural gas compressor station (GenCo supply corridor)
DBA-ES-DI-FC5	Electric	DI = Distribution	Distribution substation with edge generation
DBA-ES-CntlCo	Electric	CntlCo = coordination class	BA / RC / TOP coordination class instrument
DBA-ES-GN-FC6	Electric	GN = Generation	Coal-fired generation (>750 MW or must-run)
DBA-ES-GN-FC7	Electric	GN = Generation	Oil-fired generation and peakers (>100 MW or RMR)
DBA-ES-GN-FC8	Electric	GN = Generation	Renewable generation and BESS
DBA-DC (unchanged)	Electric / IT	DC = Data Center	Hyperscale data center (>100 MW IT load)
DBA-GAS-Sector_Framework	Natural Gas	SF = Sector Framework	Natural gas sub-sector parent

DBA-GAS-FC1	Natural Gas	—	Natural gas transmission
DBA-OIL-Sector_Framework	Petroleum	SF = Sector Framework	Petroleum sub-sector parent
DBA-OIL-FC1	Petroleum	—	Petroleum refining
DBA-OIL-FC4	Petroleum	—	Crude and product transport — trunk and product pipelines (SDD metric)
DBA-GAS-FC-LNG	Natural Gas	LNG (mnemonic)	LNG terminals — liquefaction, storage, and marine transfer (promoted from DBA-GAS-UC-LNG)
DBA-OIL-FC5	Petroleum	—	Marine crude and product terminals — strategic import/export (carved from FC3)

Section 5 — Cross-Sector Dependencies Beyond Gas-Electric

5.1 Water Sector Dependency

The water-energy interdependency is bidirectional. The forward direction runs from the energy sector to the water sector: water treatment plants, pumping stations, and distribution infrastructure are significant electric loads whose continuous operation depends on grid supply. Loss of electric supply to a major water utility is a Tier 3 consequence in the DBA-ES series, because the downstream life-safety cascade — loss of potable water supply to hospitals, dialysis centers, and dense residential populations — constitutes a cross-sector consequence chain of the severity that defines Tier 3. DBA-W §8.2 establishes dialysis centers as the most time-critical dependent facility class.

The reverse direction runs from the water sector to the energy sector: thermoelectric generating facilities — nuclear, natural gas, and coal — withdraw large volumes of water for cooling. A sustained water supply disruption affecting a power generation facility's cooling water intake can force the facility offline, compounding any grid reliability event. The formal governance boundary for this interdependency is established in Section 5.4.

5.2 Data Center Dependency

Hyperscale data centers are addressed in the DBA-DC series, which is an Electric sector sub-series instrument. Data centers are electricity consumers — not electricity infrastructure — per NIPP taxonomy. Their classification in the Electric sector design basis series reflects their consequence chain, not their sector membership. The WM-3 workload migration cross-interconnection load transient is the governing cross-sector cascade event for the data center class and is addressed in DBA-DC-L1 and DBA-ES-Sector_Framework.

The data center series is the founding member of the Electric sector's Critical Loads subsector (DBA-ES-CL), defined in DBA-ES §8.5. The forward-only elevation relationship described above is the first instance of a subsector-wide pattern rather than a property unique to data centers; subsequent Critical Loads members may carry a reverse energy-sector dependency.

5.3 Nuclear Sector Grid Dependency

Nuclear generating facilities are significant contributors to the bulk electric system. The Zaporizhzhia scenario demonstrated that military action against a nuclear facility is simultaneously a nuclear safety event and a grid reliability event. The DBA-MA nuclear series addresses the nuclear safety consequences; the DBA-ES series addresses the grid reliability consequences of nuclear facility loss. The boundary is defined in the DBA-MA Introduction §6.3. The NERC regulatory instrument governing the interface between nuclear plant operations and the bulk electric system is NUC-001-4 (Nuclear Plant Interface Coordination). NUC-001-4's NPIR coordination mechanism is the existing regulatory pathway through which nuclear-facility-

specific grid reliability requirements are made binding on Transmission Operators, Balancing Authorities, and Reliability Coordinators.

5.4 DBA-EN-SF / DBA-W Boundary Protocol

This section establishes the formal governance boundary between DBA-EN-SF and DBA-W for the water-energy interdependency. The boundary protocol addresses three questions: (a) instrument assignment for the energy supply consequence chain; (b) cross-sector consequence elevation obligation; and (c) the inheritance obligation running from DBA-EN-SF to the DBA-ES and DBA-W series.

Instrument assignment — energy supply consequence chain. A DBA-ES facility-class instrument governs the energy supply consequence chain at any facility whose loss deprives water sector facilities of their primary electric power supply. The DBA-ES instrument for the relevant facility class carries the Design Basis obligation for the electric supply consequence, including the Tier 3 designation for water sector life-safety cascades. The DBA-W series governs the water sector consequence chain at facilities within its own series scope.

Cross-sector consequence elevation obligation — forward direction (energy to water). A DBA-EN energy asset that is the sole electric supply path for a water sector facility carrying a Tier 2 or Tier 3 consequence designation in the DBA-W series must carry tightened acceptance criteria for that supply relationship, consistent with the Cross-Sector Consequence Elevation Rule in §3.4.

Cross-sector consequence elevation obligation — reverse direction (water to energy). A DBA-W water sector facility that is the sole cooling water supply path for a thermoelectric generation facility carrying a Tier 2 or Tier 3 consequence designation in the DBA-ES series must carry tightened acceptance criteria for that supply relationship in the DBA-W instrument governing the relevant facility class.

Inheritance obligation. The DBA-ES series inherits from this section the obligation to identify water sector cross-sector dependencies in each facility-class instrument and to apply the forward-direction elevation rule. The DBA-W series inherits the obligation to apply the reverse-direction elevation rule. Until the reverse-direction obligation is incorporated into each DBA-W facility-class instrument, each DBA-ES FC instrument must note the water sector cross-sector dependency as an open item in its gaps and recommendations section, consistent with DBA-ES-HY-FC1 §4.5 and DBA-ES-TX-FC2 §4.5.

Section 6 — Development Status and Open Items

6.1 Current Development Status

Current versions of every instrument named in this section are stated in DBA-EN-REG, which is the authoritative register for this series.

The Electric sub-sector series (DBA-ES) is the most developed of the three sub-sector series. Nine facility-class instruments, one functional-class instrument, one policy companion and two sector-level MA companions are active: DBA-ES-HY-FC1, DBA-ES-TX-FC2, DBA-ES-TX-FC3, DBA-ES-GC-FC4, DBA-ES-DI-FC5, DBA-ES-CntlCo, DBA-ES-GN-FC6, DBA-ES-GN-FC7, DBA-ES-GN-FC8, DBA-DC, DBA-ES-POL, DBA-ES-MA-SF, and DBA-ES-MA-QB. All FC instruments carry the full DB→DBA→DBA-MA three-layer structure, including the MA applicability four-filter gate owned by DBA-MA-AF and the CDB-1 through CDB-7 communications design basis owned by DBA-MA-CDB.

The Natural Gas sub-sector series (DBA-GAS) comprises its sector framework and eight active instruments: DBA-GAS-FC1, DBA-GAS-FC2, DBA-GAS-FC3, DBA-GAS-FC4, DBA-GAS-FC-LNG (promoted; annex DBA-GAS-UC-LNG), DBA-GAS-POL, and DBA-GAS-MA-SF. The series has completed the full three-layer architecture for the transmission and processing classes; LNG terminals have been promoted from use-case status to a standing facility class (DBA-GAS-FC-LNG).

The Petroleum sub-sector series (DBA-OIL) comprises its sector framework and six active instruments: DBA-OIL-FC1, DBA-OIL-FC2, DBA-OIL-FC3, DBA-OIL-FC4, DBA-OIL-FC5 (promoted), and DBA-OIL-MA-SF. The SDD metric is now established at both the FC4 level and sanctioned at this parent document level in §4.4. Strategic marine terminals have been carved from FC3 and registered as a standing facility class (DBA-OIL-FC5).

6.2 Open Items

OI-1: CLOSED. DBA-ES-GC-FC4 / DBA-GAS-FC1 scope boundary coordination. Resolved by §3.4.

OI-2: CLOSED. GenCo facility classes for coal, oil-fired, and renewable generation types. Resolved by DBA-ES-GN-FC6, FC7, FC8 v0.1 issuance.

OI-3: Partially addressed. Upstream and midstream petroleum facility classes. DBA-OIL-FC2 (offshore production), DBA-OIL-FC3 (terminals and storage), and DBA-OIL-FC4 (transport pipelines) are established instruments. Onshore upstream production has not yet been scoped.

OI-4: Partially addressed. Natural gas distribution and storage facility classes. DBA-GAS-FC3 (underground storage) and DBA-GAS-FC4 (local distribution) are established instruments and are reference-conformed to DBA-MA-AF and DBA-MA-CDB. Neither instantiates the four-filter applicability gate or the CDB-1 through CDB-7 communications design basis; both record the instantiation as deferred pending a gas-sector MA companion, and that companion now

exists. Instantiation is a scoping decision for the gas series with no remaining dependency, sequenced under OI-CDB-5. The residential heating dependency is addressed from the gas sector perspective.

OI-5: CLOSED. DBA-EN-SF / DBA-W boundary. Established in Section 5.4.

OI-6: CLOSED. DBA-EN-UC-2 registered and issued as v0.1.

OI-7: Open. Conflicting acceptance criteria at a dual-scope facility. The acceptance criteria assignment rule at §3.3 establishes which instrument's criteria are binding for which consequence tier. It does not establish what governs where two instruments' criteria cannot both be satisfied at the same physical asset. No DBA-EN instrument answers this. It is recorded here rather than answered, because it requires a worked case rather than a rule stated in advance. Resolution: next substantive revision of this document following a case that presents it.

Section 7 — References

Regulatory and Policy References

Department of Homeland Security. National Infrastructure Protection Plan 2013: Partnering for Critical Infrastructure Security and Resilience. Washington, D.C.: DHS, 2013.

Presidential Policy Directive 21 (PPD-21): Critical Infrastructure Security and Resilience. Washington, D.C.: The White House, February 2013.

Federal Energy Regulatory Commission. Order No. 693: Mandatory Reliability Standards for the Bulk-Power System. Washington, D.C.: FERC, 2007.

North American Electric Reliability Corporation. CIP-014-3: Physical Security. Atlanta: NERC. Current version.

North American Electric Reliability Corporation. NUC-001-4: Nuclear Plant Interface Coordination. Atlanta: NERC. Current version. Governs coordination between Nuclear Plant Generator Operators and Transmission Entities for the development and maintenance of Nuclear Plant Interface Requirements (NPIRs). Cited in §5.3 (Nuclear Sector Grid Dependency) as the NERC regulatory instrument for the nuclear-grid operational interface.

Transportation Security Administration. Pipeline Security Directive SD-02C: Cybersecurity Mitigation Actions. Washington, D.C.: TSA. Current version.

Pipeline and Hazardous Materials Safety Administration. 49 CFR Part 192: Transportation of Natural and Other Gas by Pipeline. Washington, D.C.: PHMSA. Current version.

Federal Energy Regulatory Commission, North American Electric Reliability Corporation, and Pipeline and Hazardous Materials Safety Administration. “The February 2021 Cold Weather Outages in Texas and the South Central United States.” Washington, D.C.: FERC/NERC/PHMSA, November 2021.

DBA-EN Series Instruments

Series instruments are cited in this section without version tokens. The current version of each is stated in DBA-EN-REG, which is the authoritative register for this series; a version-bearing citation here would duplicate that register and drift from it between revisions. Resolve any instrument named below against DBA-EN-REG.

Roxey, Tim. DBA-ES: Electric Sector Design Basis Framework. Eclectic Technologies.

Roxey, Tim. DBA-ES-HY-FC1: Large Hydroelectric Storage Facilities. Eclectic Technologies.

Roxey, Tim. DBA-ES-TX-FC2: Large EHV Transmission Substation. Eclectic Technologies.

Roxey, Tim. DBA-ES-TX-FC3: HVDC Interconnect. Eclectic Technologies.

Roxey, Tim. DBA-ES-GC-FC4: Natural Gas Compressor Station. Eclectic Technologies.

Roxey, Tim. DBA-ES-DI-FC5: Distribution Substation with Edge Generation. Eclectic Technologies.

Roxey, Tim. DBA-ES-CntlCo: Electric Sector Control-Company Coordination Class. Eclectic Technologies.

Roxey, Tim. DBA-ES-GN-FC6: Coal-Fired Generation. Eclectic Technologies.

Roxey, Tim. DBA-ES-GN-FC7: Oil-Fired Generation. Eclectic Technologies.

Roxey, Tim. DBA-ES-GN-FC8: Renewable Generation and BESS. Eclectic Technologies.

Roxey, Tim. DBA-GAS-Sector_Framework. Eclectic Technologies.

Roxey, Tim. DBA-GAS-FC1: Natural Gas Transmission. Eclectic Technologies.

Roxey, Tim. DBA-GAS-FC2: Natural Gas Processing. Eclectic Technologies.

Roxey, Tim. DBA-GAS-FC3: Underground Storage. Eclectic Technologies.

Roxey, Tim. DBA-GAS-FC4: Local Distribution. Eclectic Technologies.

Roxey, Tim. DBA-OIL-Sector_Framework. Eclectic Technologies.

Roxey, Tim. DBA-OIL-FC1: Petroleum Refining. Eclectic Technologies.

Roxey, Tim. DBA-OIL-FC2: Offshore Crude Production. Eclectic Technologies.

Roxey, Tim. DBA-OIL-FC3: Crude and Product Terminals and Storage. Eclectic Technologies.

Roxey, Tim. DBA-OIL-FC4: Crude and Product Transport. Eclectic Technologies.

Roxey, Tim. DBA-MA-AF: MA Applicability Framework — Four-Filter Applicability Gate. Eclectic Technologies.

Roxey, Tim. DBA-MA-CDB: Military Action Communications Design Basis. Eclectic Technologies.

Supporting Documents

Roxey, Tim. GenMix Optimization Whitepaper. Version 0.3 — DRAFT. Eclectic Technologies, 2026. (Origin of RT_HVLLC construct.)

Roxey, Tim. Predictive UFLS Ride-Through Position Paper. Version 0.3 — DRAFT. Eclectic Technologies, 2026. (PMU wavefront analysis; cross-reference DisCo PMU taxonomy entry.)

Defense Science Board Task Force on DoD Energy Strategy. “More Fight — Less Fuel.” Washington, D.C.: Office of the Under Secretary of Defense for Acquisition, Technology and Logistics, February 2008.

Revision History

Version	Date	Author	Description
0.1	May 2026	T. Roxey	Initial working draft. Established Energy sector parent document architecture. Defined three sub-sector series: DBA-ES (Electric), DBA-GAS (Natural Gas), DBA-OIL (Petroleum). Established an inductive architecture principle. Named the gas-electric bidirectional cascade as the governing cross-sub-sector interdependency. Confirmed DBA-ES retains its own designation and is not renamed under a DBA-EN hierarchy.
0.2	June 2026	T. Roxey	Substantive revision to serve as the authoritative parent document for the new Energy Sector project. Incorporated the complete Energy Sector taxonomy from Design Basis Energy Sector Mindmap v2 (June 2026). Added full GenCo taxonomy; TransCo infrastructure detail; DisCo control systems detail; CntlCo completeness. Added Oil and Natural Gas sub-sector taxonomy. Formalized gas-electric bidirectional cascade as a first-class cross-sub-sector postulated event category. Added sub-sector series inventory table. Added dual-DB architecture acknowledgment. Added Bright Line and Command Broker as cross-sector series requirements. Added series-wide naming convention and references section.
0.3	June 2026	T. Roxey	Section 3.4 (Cross-Sector Consequence Elevation Rule) added. Establishes the governing coordination protocol for DBA-ES-GC-FC4 and DBA-GAS-FC1 at dual-scope compressor station facilities: consequence-perspective separation rule; acceptance criteria assignment rule; and cross-sector consequence elevation rule. Names the NERC CIP gap. Adopts the single-path dependency criterion from FC4 §1.2. Section 5.3 (Nuclear Sector Grid Dependency) expanded; NUC-001-4 added. OI-1 closed.
0.4	June 2026	T. Roxey	Section 5.1 (Water Sector Dependency) expanded to address the bidirectional water-energy interdependency. Section 5.4 (DBA-EN-SF / DBA-W Boundary Protocol) added. Establishes instrument assignment rule, forward-direction cross-sector consequence elevation obligation, reverse-direction elevation obligation, and inheritance obligations. OI-5 closed.
0.5	June 2026	T. Roxey	Section 3.4 (Cross-Sector Consequence Elevation Rule) revised to implement the bidirectional rule fully. Named-series elevation trigger table added: DBA-W (bidirectional), DBA-DC (forward only), DBA-MA nuclear (bidirectional). The notification protocol paragraph has been established. Open items OI-DC-1 and OI-MA-1 registered. Gap 1.3 closed at the

			DBA-EN-SF level.
0.6	June 2026	T. Roxey	Section 4.4 (Consequence Metrics) revised: opening sentence updated from “Three” to “Four” consequence-duration metrics. PPSD (Pipeline Pressure Supply Duration) was added as the fourth series-standard metric. Definition per DBA-GAS-FC2 §1.3; PPSD→GSD sequential relationship established by reference to DBA-EN-UC-1 §5.1. Gap 1.6 closed.
0.7	June 2026	T. Roxey	Section 4.4 (Consequence Metrics) revised: opening sentence updated from “Four” to “Six” consequence-duration metrics. SDD (Supply Delivery Duration) added as the fifth series-standard metric: governing metric for DBA-OIL-FC4 crude and product pipeline transport facility class; established in DBA-OIL-FC4 v0.3 §§1.3 and 4.2; non-additivity rule stated (governing SDD is the longest single-segment critical path); Zone 1/2/3 acceptance criterion framework with T_1/T_2 calibration noted. SABD (Situational Awareness Blackout Duration) added as the sixth series-standard metric: governing metric for DBA-ES-CntlCo functional class; established in DBA-ES-CntlCo v0.2 §1.2; BCC transition non-termination rule stated; Zone 1/2/3 escalation structure with T_1/T_2 calibration noted. Both metrics follow the same sanctioning pathway as PPSD (established at FC level, sanctioned upward to DBA-EN-SF).
0.8	June 2026	T. Roxey	Housekeeping revision. §1.4 Sub-Sector Series Inventory table refreshed across two passes. Initial pass (v0.8 baseline): GC-FC4 corrected to v0.4, DI-FC5 corrected to v0.3, DBA-DC corrected to v1.7, DBA-ES-MA-SF v0.2 added to DBA-ES row; OIL-FC4 corrected to v0.3; DBA-EN Cross-Sector row added (UC-1, UC-2, MA-SF, DBA-VC v0.3). Second pass (N-15/N-16 batch pass closure): all nine DBA-ES FC instrument versions updated to post-batch-pass current versions (HY-FC1 v0.5, TX-FC2 v0.3, TX-FC3 v0.3, GC-FC4 v0.5, DI-FC5 v0.4, CntlCo v0.3, GN-FC6 v0.2, GN-FC7 v0.2, GN-FC8 v0.2); DBA-ES-MA-QB (v0.1) added to DBA-ES row; DBA-ES-MA-SF removed from DBA-EN cross-sector row (ES-scoped instrument, correctly listed in DBA-ES row only). Body text references updated to match throughout. §10 series narrative updated to reflect current instrument count and versions. Closes Gap N-1 in DBA-EN Gaps and Recommendations v0.4.
0.9	June 2026	T. Roxey	Parent-registration pass for the marine/strategic terminal promotions and the natural gas and petroleum MA-SF companion layer. §1.4 inventory: DBA-GAS row updated to

			v1.7 with DBA-GAS-FC-LNG and DBA-GAS-MA-SF (v0.2) added; DBA-OIL row updated to v1.8 with DBA-OIL-FC5 and DBA-OIL-MA-SF (v0.1) added; DBA-ES Sector Framework version corrected v0.8 → v0.9. §4.5 naming convention: promoted-class naming clause added (next sequential number when carved from an existing class; use-case mnemonic when promoted from a named use case); DBA-GAS-FC-LNG and DBA-OIL-FC5 registered. §6.1 Development status refreshed for the gas and petroleum series. Registers the promotions asserted in DBA-GAS-MA-SF v0.2 / DBA-OIL-MA-SF v0.1 and the Sector Framework registrations in DBA-GAS-SF v1.7 / DBA-OIL-SF v1.8.
0.10	June 2026	T. Roxey	Critical Loads subsector (DBA-ES-CL) registered. §1.4 inventory: DBA-DC grouped under the Critical Loads subsector (DBA-ES-CL) within the DBA-ES row; DBA-ES Sector Framework updated v0.9 → v0.10 and DBA-DC updated v1.7 → v1.9 to reflect the instruments issued in this pass. §4.5: note added that a sub-sector may organize facility classes into named subsector groupings (DBA-ES-CL), which do not alter instrument designations. §5.2: DBA-DC identified as founding member of DBA-ES-CL. §6.1 Development status refreshed (DBA-ES v0.10, DBA-DC v1.9). The Critical Loads subsector is defined in DBA-ES §8.5.
0.11	Septe mber 2026	Tim Roxey	Reference conformance to DBA-MA-CDB and DBA-MA-AF. The series status note re-points the MA applicability gate to DBA-MA-AF and the communications design basis to DBA-MA-CDB, which states it as CDB-1 through CDB-7 and supersedes the CB-1–CB-5 enumeration. The running-header version token, which had lagged at v0.8, is corrected to match the cover. No requirement in this instrument changes.
0.12	Septe mber 2026	Tim Roxey	Register reconciliation and reference-convention change. §1.4 Sub-Sector Series Inventory reconciled to the instruments on record. DBA-ES row: HY-FC1 v0.6, TX-FC2 v0.4, TX-FC3 v0.4, GC-FC4 v0.6, DI-FC5 v0.5, CntlCo v0.5, GN-FC6/FC7/FC8 v0.3, MA-SF v0.3, MA-QB v0.2. DBA-GAS row: FC3 v0.3, FC4 v0.3. DBA-OIL row: FC2 v0.3, FC3 v0.3. DBA-EN row: this document's own Current Version cell corrected from v0.9; DBA-VC corrected v0.3 → v0.9; shared machinery modules DBA-MA-CDB (v0.3) and DBA-MA-AF (v0.2) registered as active cross-sector instruments. CntlCo, GAS-FC3, GAS-FC4, OIL-FC2 and OIL-FC3 were already behind the register before the DBA-MA-CDB conformance pass; the register is the governing statement under the reconcile-

			to-§1.4 rule, so its lag propagated. Statements of current state elsewhere in the document reconciled to §1.4: ten §2 taxonomy treatment cells, the three §6.1 development-status paragraphs, and open items OI-3 and OI-4 (OI-3 and OI-4 restated — GAS-FC3 and GAS-FC4 are no longer at first issue, and their non-instantiation of the gate and the communications design basis is a scoping question rather than a blocked dependency). §7 References: DBA-EN series instruments are now cited without version tokens, with a lead-in note directing readers to §1.4. A version-pinned citation to an instrument in this series is a pointer into the register, not a dated bibliographic fact; pinning it recreates the defect recorded as AF-OI-9, where citations naming a superseded version resolved to superseded text sitting on disk beside its successor. External regulatory references and supporting documents are unchanged. DBA-MA-AF and DBA-MA-CDB added to the series instrument list. Closes OI-CL-2. No requirement in this instrument changes.
0.13	September 2026	T. Roxey	§1.4 Sub-Sector Series Inventory: the DBA-MA-CDB entry annotated “canonical home DBA-MA parent,” matching the annotation the DBA-MA-AF entry already carried. Records the transfer of the DBA-MA-CDB master to the DBA-MA parent series, PQ 9 having been closed by DBA-MA-DN v0.2. No requirement is changed.
0.14	September 2026	T. Roxey	§1.4 Sub-Sector Series Inventory, version-only touch. DBA-MA-CDB carried v0.3 → v0.4 and DBA-MA-AF carried v0.2 → v0.3, both issued at the DBA-MA parent on 2 September 2026 — CDB v0.4 restating §4 with the reconciliation condition recorded as satisfied by DBA-MA-DN v0.2 and the migration recorded as performed; AF v0.3 restating the AF-MIGRATION NOTE with the deferral discharged and the canonical home recorded as the parent. The canonical-home annotations on both entries are unchanged. The register records what exists. No requirement in this instrument changes.
0.15	September 2026	T. Roxey	§1.4 Sub-Sector Series Inventory reconciled to the seventeen instruments issued under OI-R3-1 on 2 September 2026, and §4.3 given the propagation closure. §1.4: DBA-ES row — TX-FC2 v0.5, TX-FC3 v0.5, GC-FC4 v0.7, DI-FC5 v0.6, CntlCo v0.6, DBA-DC v1.11, GN-FC6/FC7/FC8 v0.4; DBA-GAS row — Sector Framework v1.8, FC1 v1.7, FC3 v0.4, FC4 v0.4; DBA-OIL row — FC1 v1.5, FC2 v0.5, FC3 v0.4; DBA-EN row — this document to v0.15. This lifts the suspension of the reconcile-to-§1.4 rule that Governing Instructions v0.18 §1 placed on those rows. §4.3: a closing paragraph states that sub-

			sector frameworks and facility-class instruments inherit the placement criterion by reference to §4.3 and shall not restate it, with OI-R3-1 named as the reason. The closure previously sat at DBA-DC-L1 §3.5, one level below the section it governs. NOTE — the OI-R3-1 pass cited this rule as "DBA-EN-SF §4.2" in thirty body sites across seventeen instruments; §4.2 is Dual Design Basis Architecture and the placement criterion is and was §4.3. Those citations are corrected in the version-pin pass, not here. NOTE — forty version-pinned in-series mentions elsewhere in this document (the sub-sector coverage table and the three development-status paragraphs) now name superseded versions and are NOT reconciled by this touch; they are the in-series citation convention's business and go to the same pass. No requirement is changed.
0.16	September 2026	T. Roxey	Internal register reconciliation. The §2 sub-sector taxonomy treatment cells, the three §6.1 development-status paragraphs and the §1 lineage sentence are reconciled to §1.4, which stands as issued at v0.15 and is unchanged by this touch. Twenty-seven sites: ten in §2 (TX-FC2 v0.5, TX-FC3 v0.5, DI-FC5 v0.6, GC-FC4 v0.7, GAS-FC1 v1.7, GAS-FC3 v0.4, GAS-FC4 v0.4, OIL-FC1 v1.5, OIL-FC2 v0.5, OIL-FC3 v0.4); sixteen in §6.1 (the same instruments plus CntlCo v0.6, GN-FC6/FC7/FC8 v0.4, DBA-DC v1.11, and the DBA-GAS sector framework stated at v1.8); and the §1 sentence, which read that this document is drafted at v0.3, recast to record that it was first drafted at v0.3. The two §4.4 metric provenance anchors — SDD at DBA-OIL-FC4 v0.3 and SABD at DBA-ES-CntlCo v0.2 — are left as written, because they record where a metric was established and do not go stale. Eleven further in-series mentions were pinned and already current; they are left pending application of the in-series citation convention to this document. REASON. Under CR-CF §5 the bare-citation form is available to this series because §1.4 is a register to hold the versions the citations give up. A bare citation resolving into a document that carries two registers disagreeing with each other is worse than the stale pin it replaces, because the pin at least shows a reader a version to doubt. The disagreement is closed before the facility-class instruments inherit the version-free lead-in. Raised by Project 6 on 3 September and read site by site rather than screened: the proximity screen found twenty-six and reading found a twenty-seventh, where the version token does not sit adjacent to the designation. Cover, running header, footer, filename, the §1.4 self-version cell and this row all move together (OI-REV-1). No requirement in this instrument changes.

0.17	September 2026	T. Roxey	Requirement change and citation-strip pass, and they are different in kind. REQUIREMENT. §3.3 gains the acceptance criteria assignment rule, which the v0.3 revision row announced as one of three rules established that cycle and which was in the body of neither §3.3 nor §3.4. It is promoted rather than drafted: DBA-ES-GC-FC4 states it in its DI-3 closure and DBA-GAS-FC1 applies it, and both cite §3.3 for it, so this touch makes those citations resolve. DBA-ES-GC-FC4 states its DI-3 closure ground five times, four carrying the full §3.3-and-§3.4 pairing and one §3.4 alone; the single narrow statement is an error for correction at that instrument's next touch and was never a second account with standing. This is the first requirement change to this instrument since v0.11, and the "no requirement changes" formula of v0.15 and v0.16 does not apply. §6.2: OI-7 opened for the case the new rule does not close — two instruments' criteria irreconcilable at one physical asset — recorded rather than answered because it needs a worked case. OI-4 restated on a first-hand read of both gas instruments: each records in terms that it does not yet carry the gate or CDB-1 through CDB-7, so the claim stands, but the deferral's stated ground has gone — the gas-sector MA companion it waits on now exists. CITATION STRIP. Forty-two in-series version tokens removed under the D1 determination at Version_Pin_Pass_Scoping_Note: twelve in §2 treatment cells, twenty-eight in the three §6.1 development-status paragraphs, two in §6.2 OI-4. §6.1 gains one pointer sentence to §1.4 rather than three. §1.4 keeps its versions — it is the register. The §4.4 metric-provenance anchors, the historical narrative in §1.3, §3.4, OI-2 and OI-6, and every revision-history row are out of scope and untouched. COUNT. The pass was scoped at fifty-nine sites by screen; reading returns forty-two in scope out of one hundred and eighty-one version tokens in the document. The forty this document's own v0.15 row named are exactly the §2 twelve plus the §6.1 twenty-eight. Screen to find, read to count. §6.1 CORRECTION. Its first sentence counted "eleven facility-class instruments, one functional-class instrument, and one sector-level MA companion" and then named thirteen; the total was right and the categories were not. Corrected to nine facility classes, one functional class, one policy companion and two MA companions.
0.18	October 2026	T. Roxey	Conformance revision ahead of re-publication, on Project 0's split ruling of 27 September 2026. No requirement changes. §1.4 becomes a pointer to DBA-EN-REG, as that register's §3 provided at its issue on 9 September; its inventory table is removed and remains in v0.17. The series lead-in at §6.1 and §7

			now names DBA-EN-REG as the authoritative register instead of §1.4. Facility-class type conformed under Project 0's determination of 30 September 2026: CntlCo is a coordination class, at five body sites and two cells of the §4.5 naming-convention table, whose column heading becomes Facility Class. The §7 reference to DBA-ES-CntlCo carries that instrument's new title.
0.19	November 2026	T. Roxey	Requirement change. Section 3 is retitled Cross-Sub-Sector Postulated Events, and §3.5 is added. It names the coordinated multi-node attack as the second first-class cross-sub-sector postulated event, beside the gas-electric cascade, across all three sub-sector series. It states the event, its consequence and why single-contingency design, including the N-1 criterion, is no defense, and cites DBA-EN-UC-2 for the method without restating it. Facility-class instruments inherit the obligation by reference at their next substantive revision; no open item is opened. Taken on the author's acceptance on 7 November 2026 of Project 10's note of 3 November 2026. §3.1–§3.4 are unchanged and keep their numbers. OI-6 stands as written. The cover date, which had stood at June 2026 since an early version, moves to November 2026. Edition 1 of CIKR Design Basis Examples keeps v0.18; this version enters the next edition.